

MOVIE RECOMMENDATION USING MACHINE LEARNING

Submitted By:

Prachi Netke

Department of Artificial Intelligence & Machine Learning

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INTRODUCTION

A Movie Recommendation System is an important application of Machine Learning that helps users discover movies based on their interests and preferences. It analyzes user behavior, ratings, and movie information to suggest relevant movies.

With the increasing number of movies available on online platforms, it becomes difficult for users to choose what to watch. A recommendation system provides personalized suggestions, improving the user experience and saving time.

This project aims to develop a Machine Learning model that recommends movies based on user preferences and movie features. The system uses movie datasets and recommendation algorithms to predict movies that a user is likely to enjoy.

The insights obtained from this project can improve user satisfaction, increase engagement on streaming platforms, and provide a better entertainment experience.

PROBLEM STATEMENT

Users often face difficulty in selecting movies from a large collection available on streaming platforms. Searching manually is time-consuming and may not always provide suitable results.

By analyzing user ratings, movie genres, and viewing history, Machine Learning can recommend movies that match user interests. This helps users quickly find relevant movies and improves their overall experience.

OBJECTIVES

The primary objectives of this project are:

To analyze movie and user rating data.

To preprocess and clean the movie dataset.

To perform Exploratory Data Analysis (EDA) on movie data.

To build a Machine Learning-based recommendation system.

To recommend movies based on user preferences.

To improve recommendation accuracy and user satisfaction.

TECHNOLOGIES USED

The following technologies and tools were used for the development of the Movie Recommendation System:

Python

Programming language used for model development.

Google Colab

Cloud-based platform used for coding and execution.

Pandas

Used for data manipulation and analysis.

NumPy

Used for numerical computations.

Matplotlib

Used for data visualization.

Seaborn

Used for statistical data visualization.

Scikit-learn

Used for machine learning model development.

Machine Learning Techniques Used

Content-Based Filtering

Collaborative Filtering

Cosine Similarity

DATASET DESCRIPTION

Dataset Name: MovieLens Dataset

Total Records: 100,000+ Ratings

Movies: 1,600+

Users: 943+

Target: Recommend movies based on user preferences

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